Addressing the Gender Digital Divide to Improve Income Generation in Ransai Village, Maharashtra

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***Abstract*— The digital divide in rural communities, particularly among women, exacerbates socio-economic inequalities and hinders income generation. This paper examines the impact of the gender digital divide in Ransai village, Maharashtra, and proposes solutions to empower women through digital literacy. Using participatory design, we identified key gaps in digital skills among women and propose community-driven workshops to empower them. Our findings suggest that addressing this digital divide can improve income-generating opportunities for women and foster sustainable development in the village.**

***Keywords*— gender digital divide, digital literacy, participatory design, sustainable development, rural economy, Ransai village**

I. Introduction

The digital divide is there in three levels [1]. First level is digital access divide [2], which denotes the disparities in accessing the internet. Second level is digital usage or capability divide [3], which means the differences in the internet usage and digital skills of the users. The third level is the digital outcome divide [4], focusing on the internet benefits to attain sustainable development goals. Digital access divide is more in rural communities compared to urban communities due to low population density. High Capital Expenditure (CAPEX) and Operational Expenditure (OPEX) are making it difficult for the telecommunication service provider to provide a viable business model in rural communities [5].

The Ransai village, located in Khalapur Taluka, Raigad district of Maharashtra [6], has a population of approximately 3,000 people across six localities, or "wadis". The village is pre-dominantly agrarian, with farming serving as the primary source of income [7]. The population's socio-economic structure is defined by traditional family hierarchies and agricultural cycles. However, this close-knit community lacks modern digital infrastructure, which limits their access to education, economic participation, and technological advancements. Being situated in a hilly terrain region, Ransai faces significant challenges in cellular connectivity implementation. The mountainous topography makes it particularly difficult for service providers to establish and maintain reliable network infrastructure. Moreover, the relatively small population and challenging geography make it economically less viable for telecommunication companies to invest in comprehensive cellular coverage, as they cannot expect a substantial user base to justify the high implementation costs in such challenging terrain [8]. Ransai village one year back only solved digital access divide to few houses by setting local Wi-Fi access through local internet service provider. There is digital divide among the poor, poorer and poorest sections of households in this tribal village, like other rural communities in India.

Gender digital divide [9] is also evident among the poorest and poor sections in Indian rural communities like Ransai. The global mobile gender gap report also presents their findings as only 19% of women is less likely than men to use mobile internet and 17% of women is less likely to own a smartphone than men [10]. While men generally have greater access to mobile phones and the internet, women are largely excluded from these technologies due to cultural norms, lack of education, and economic barriers. Women's access to technology remains limited, curtailing their opportunities to participate in income-generating activities like e-commerce, digital banking, or accessing government schemes online. This digital exclusion also prevents women from gaining digital literacy, further deepening socio-economic inequalities in the community. This community-driven challenge necessitated the use of Participatory Rural Appraisal (PRA) tools, which is an authentic framework for data collecting and analysis, which were familiarized and guided by the Live in Labs team.

The Live-in-Labs® program at Amrita Vishwa Vidyapeetham is a pioneering, multidisciplinary experiential learning initiative aimed at fostering sustainable rural development in India [11]. Integrating academic theory with real-world application, the program invites students, researchers, and experts from diverse fields to engage directly with rural communities. Through immersive stays in villages, participants gain a deep understanding of the socio-economic, environmental, and infrastructural challenges these communities face [12]. By working collaboratively with locals, they co-create and implement tailored interventions, targeting areas such as health, livelihood, education, energy, and infrastructure, to promote self-reliance and sustainable income generation [13], [14], [15]. This approach, embedded in the curriculum, not only bridges academic and practical knowledge but also emphasizes participatory, human-centered solutions that reflect the community's needs, creating

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meaningful, scalable impact across India's rural landscape [16].

This paper explores the research question of how bridging the gender digital divide can enhance income generation in rural communities, using Ransai village in Maharashtra as the case study. This study describes the participatory research methodology adopted to understand the Ransai village community in Section 2. PRA tools analyzed are presented to understand the community problem in Section3. Findings about the gender digital divide problem and feasible high impact solution are illustrated in Section 4. Section 5 describes the proposed solution to bridge the gender digital divide in the rural community like Ransai.

II. Participatory Research Methodology

A. Participatory design method

An ethnographic research approach is adopted, immersing the authors to better understand the daily lives and challenges of the villagers, especially the women in the community. The methods included PRA tools, such as resource mapping, transect walks, and interviews, to identify the community's needs and challenges. The ethnographic approach enabled us to capture the local context, which is essential for designing sustainable and context-sensitive solutions pertaining to that community [16].

B. Data Collection Techniques

Data was collected through mixed methods, combining quantitative surveys and qualitative interviews to examine the research question. After taking verbal consent from the villagers, surveys and interviews were conducted and names kept anonymous to maintain privacy. The household surveys focused on digital access, ownership of devices, and internet literacy levels. Focus group discussions and brainstorming sessions with women's groups were conducted to explore their interest in learning new skills and their perceived barriers to digital engagement. Key stakeholders, including village elders and panchayat members, were interviewed to gain deeper insights into the socio-economic structure of the village.

III. Participatory Tools Analysis & Findings

Various PRA tools like resource maps, income and expenditure, problem tree are analyzed to find the opportunities and problems affecting the socio-economic growth of the village.

A. Resource Maps

The resource map was developed to identify available infrastructure and resources. The resource map of the same village is utilized to provide water facilities to the household by empowering women is mentioned in [17]. Fig. 1 is the resource map of the Ransai village designed by the Sankalp team along with the participation of the community focuses on digital access resources. It reveals that the village has no access to cellular network. But the village has limited access to the internet through Wi-Fi. These resources primarily controlled by male members of the households. Women had significantly less access to smartphones, with many relying on shared family devices. This lack of infrastructure is compounded by cultural barriers that limit women's mobility and access to internet facilities.

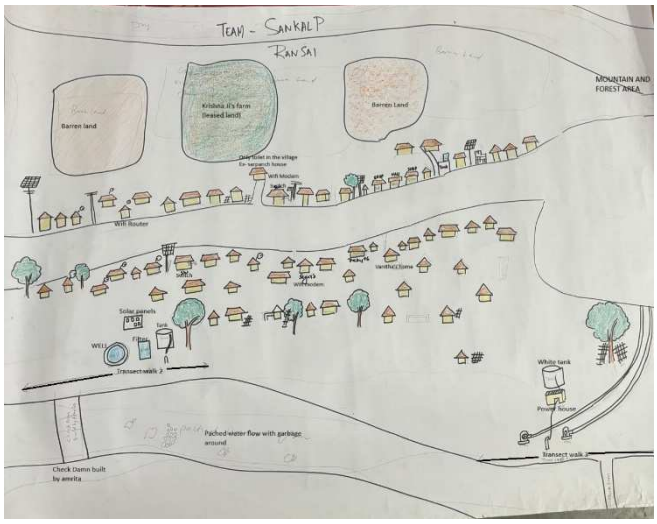


Fig. 1: Resource map of Ransai village by Sankalp team

Cellular communications are not yet received in the Ransai tribal community, due to the hilly terrain and being a remote and poor village. Due to intervention by a local male villager, internet access is deployed in the remote area as illustrated in the community network topology shown in the Fig. 2. The backhaul fiber was received from the local internet service provider Bhawani. Mumbai, being metropolitan urban areas in Maharashtra, have very good internet access. But this remote tribal village Ransai got connected to internet through Wi-Fi recently only after covid pandemic times.

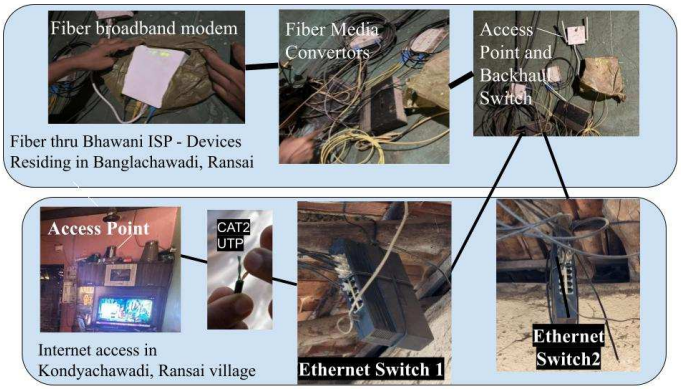


Fig. 2: Community Network Topology deployed

B. Lack of Circular Economy

The income and expenditure table are designed after the quantitative and qualitative surveys carried out in the community.

		Expenditure (in Rs)	Income (in Rs)
Energy	Electricity	350/- to 500/- per month	nil
	Petrol/Travel expenses	250/- to 500/- per month	nil
	Gas	400/-	nil
	Wifi	450/-	nil
Water	Tanks during summer	2000/-	nil
Livelihood	Faming	30,000/-	50,000/-
	Groceries	500/-	nil
	Education	~ 20,000 to 70,000	nil
	Shops	~ 250/- to 500/-	nil
	Labour work	Travel expenditure - 500/-	500
Health	Labor Charges	40,000 to 50,000/-	nil
	General Health Care expenses	5,000	nil

Fig. 3 Income and expenditure of an individual household

Thus, a significant disparity between income and expenses is obvious in Fig. 3. Most of the money leaves the village

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whereas very little comes back to it in return. Key sectors like energy, water management, and labor work lack mechanisms to reinvest or circulate money within the village. This dependence on external resources and lack of local income streams highlights the need for a circular economy [1], where local businesses, sustainable resource management, and job creation could reduce outflow and generate more internal financial resilience.

C. Problem Tree Analysis

Using Problem Tree, one of the PRA tool, the root causes of the gender digital divide in Ransai is identified after brainstorming with the community as shown in the Fig. 4. The seasonal agricultural occupation causes a lack of stable income. Geographical terrain and remote village increase the travel expenses and faces difficulty in affording good healthcare centers located in urban areas.

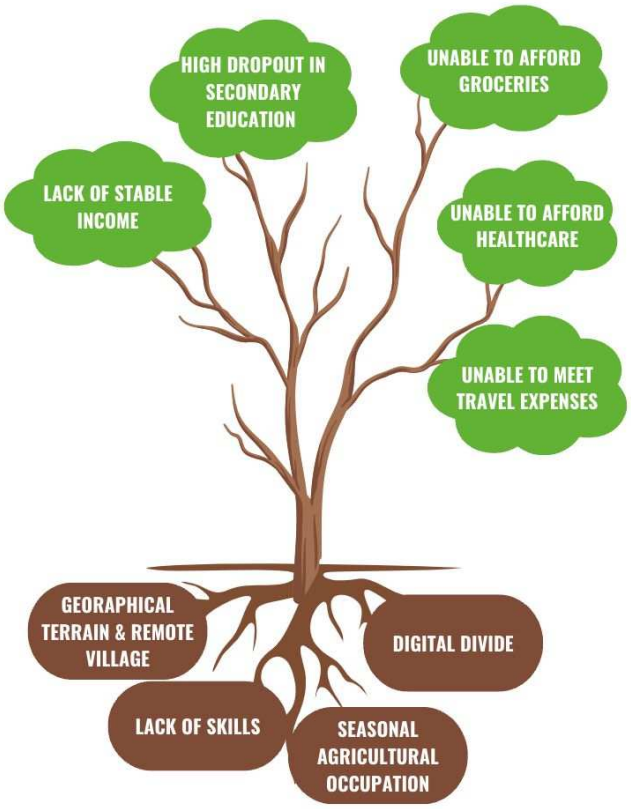


Fig. 4: The Problem Tree

The analysis revealed several critical issues:

- **Cultural Barriers:** Traditional gender roles restrict women's access to technology and limit their participation in digital spaces.
- **Access to Technology:** Women have limited ownership of digital devices such as smartphones and computers.
- **Digital Literacy:** A significant portion of women in Ransai are not familiar with using digital tools, such as mobile phones and the internet.
- **Economic Barriers:** Many women face financial barriers that prevent them from accessing digital devices or paying for internet services.

These factors collectively contribute to the marginalization of women in Ransai, inhibiting their ability to participate in income-generating activities and community development efforts through internet access and usage.

D. Findings

In Ransai Village, the community can be divided into 3 different sections based on income, electricity access, and digital resources. The 3 sections of the community are classified as poor, poorer and poorest as shown in Table I. The poorest households struggle with high transport costs, limited electricity, and education, with children rarely studying beyond 4th grade. They lack digital access, further limiting income opportunities. The poorer segment fares slightly better but still faces unstable income and limited education for girls beyond 8th grade. Meanwhile, the poor households seek to improve their standard of living through additional income. Across all sections, the gender digital divide is evident, with women having less access to digital tools, restricting their potential for income generation and economic empowerment.

TABLE I. SOCIO-ECONOMIC DIVIDE IN RANSAI COMMUNITY

Level	Socio-economic & digital divide disparities
Poorest	• Unable to afford to pay Electricity bill – Rs. 350 -500 /month. • Difficult to meet transport charges Rs. 250-500 /trip for travelling to daily labour jobs. • Difficult to even meet minimal health care expenses. • Children studying till 4th but not beyond. • Not able to afford additional groceries. • Experiences low digital access itself. • Men have ordinary cell phones without internet access and women don't have access to phone also.
Poorer	• Transport – Rs. 500/- and more per trip for travelling to daily labour jobs and to meet school studying boy kids in hostel is affordable. Girls travel back and forth from village to school. Meeting transport expenses is challenging for this section also. • Able to afford education after 8th for girls and 12th for boys. • Digital access divide(level1) is narrowed down in this section compared to Poorest section. • Men have smart phones with internet access and women share with the family members. • Women lack skills to use mobile phone belonging to male members of the family
Poor	• Improving the standard of living via extra income. • Digital access and usage divide (level2) is narrowed down compared to other section of society. • Women watches YouTube streaming in TV also. • Men additionally able to pay electricity bills, gather information about farming through internet in phones, but women are not skilled enough to reap the internet benefits – digital outcome divide (level 3)

In spite of digital access deployed, there are disparities in the internet access among the three socio-economic sections in the society along with the gender barriers in each of these three sections is more prevalent in the Table I. Men are skilful in providing internet access through Wi-Fi and stream in TVs also. Fig. 5 shows the YouTube streaming of cricket match played by players from the same villager performing in the Mumbai playground is witnessed by their family members in TV. Thereby digital resources reduces the travel expenses without compromising the Quality of Life (QoL).

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Fig. 5: YouTube Streaming in TV

It is also observed that men is skilful in using digital resources compared to women. YouTube streaming of cricket match played by their family members in local tournament in Mumbai is watched first by men and then followed by women later only. This proves the gender digital divide prevalent in village.

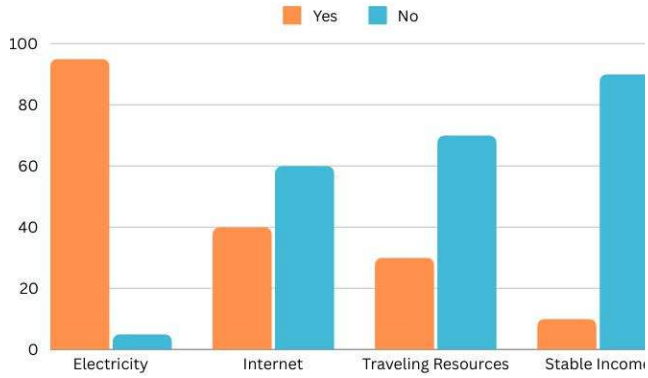


Fig. 6: Household disparities

Fig. 6 shows the proportion of Electricity, internet, travelling resources like vehicles and stable income in all the houses in the village. Before electricity came to the village, the solar panels were installed by the NGO in the community. After one phase electricity arrived in the village, most of the houses have electricity either having meter in their house or shared by the neighbour. 40% households have internet access and 30% households own travelling vehicles like motor bike and tractor mentioned in the Fig. 6 shows the challenge of meeting the travelling expense with no stable income in most of the families. Stable income is for only few dedicated households who have multiple income sources like farming, transportation vehicles for rent and shops. But most of the villagers only have seasonal income with agriculture and face the challenges of low income during summer season.

Fig. 7 show the male and female disparities with respect to owning a mobile phone, digital skills and secondary education completed level. Disparities in owning mobile phone as used by the men and son compared to the wife and daughters is very huge. The qualitative interviews with one of the active male villager revealed that he has digital skills to do online banking and reduces the travel expenses by paying the electricity bill online. There are other community members in poorer and poorest section who lack digital skills and have low income to pay for travel expenses are unable to pay the electricity bill on time. Thus, electricity is lost in their houses due to lack of digital skills in addition to having mobile phones.

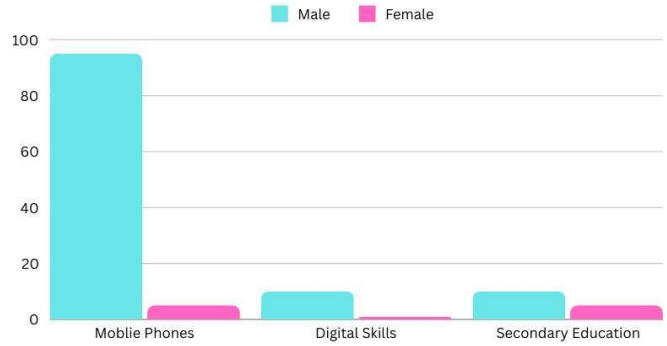


Fig. 7: Gender disparities in Ransai village

One interviewed women who had completed her secondary education and going to college owns a smartphone and has basic digital skills. Other interviewed women is a village Panchayat representative, owns a smart phone with internet access to send reports to the government. Thus, education and occupation are driving factors for women to bridge the gender digital divide. In the focus group discussions with women in the community, they were requesting for skills training that can support them to earn the income for their family. Thus, one male bread winner can be supported by additional female bread winner in the same family to support the dependents.

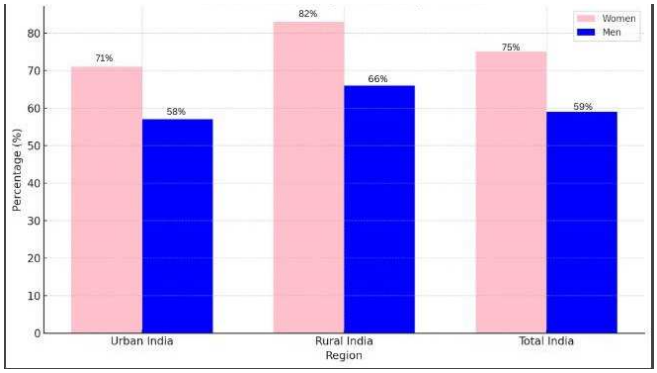


Fig. 8: Gender based digital illiteracy in India

The graph shown in Fig. 8 highlights the digital illiteracy rates in India based on data from the National Family Health Survey (2019-21) [19]. It shows a significant gender and rural-urban divide in internet access. Urban women have a digital illiteracy rate of 59%, compared to 43% for urban men. Rural women have the highest digital illiteracy rate at 75%, while rural men have a rate of 51%. This reveals a wider digital gender divide in rural areas compared to urban areas. This indicates that men in urban areas are more digitally literate than women. The national data comprises of 0.2 million Indians from all states and union territories underscores the need for targeted initiatives to improve digital access and literacy, particularly for women in rural areas, to bridge the digital gender divide and ensure inclusive growth in India's digital economy.

The solutions to bridge the gender digital divide for rural communities is evaluated as shown in the Fig. 9. Building schools and hospitals can have high societal impact, but it is hard to implement. Creating formal employment opportunities is hard to implement and will have low impact, if community

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members are not motivated. The easy implementation with low societal impact is just giving awareness campaigns on digital literacy, which will not be effective to make a positive change in the community. The easy implementation with high impact can be considered as empowering women self-help groups to initiate and manage local business.

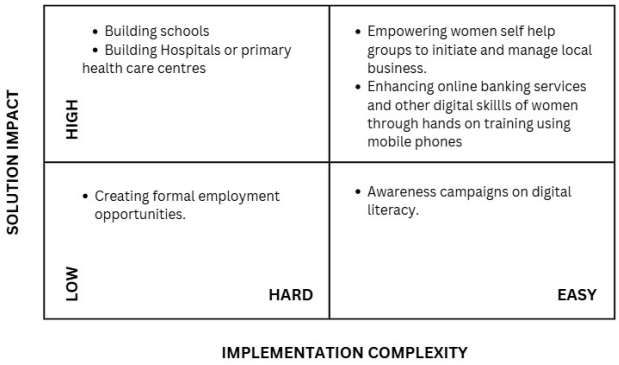


Fig 9: Solution complexity analysis

IV. Proposed Solution

A. Co-Design Workshops for Skill Development

Based on the findings from our research, we initiated co-design workshops with the women of Ransai. The aim was to involve them in the process of designing digital literacy programs that meet their needs and interests. During these workshops, women expressed a strong interest in learning how to use mobile phones for e-commerce, digital banking, and accessing government schemes. However, they also highlighted their lack of confidence in using digital devices.

B. Empowering a Community Leader

To ensure sustainability, we propose training a local woman leader who can act as a peer educator and digital ambassador within the community. This woman would undergo intensive digital literacy training and would be responsible for teaching other women in the village. By empowering a local leader, we ensure that the skills are passed on in a culturally acceptable manner, encouraging more women to participate in the digital literacy program.

C. Digital Literacy Workshops

We recommend the implementation of a series of digital literacy workshops aimed at:

- Teaching women basic mobile phone usage, such as internet navigation, digital payments, and online marketplaces.
- Providing them with access to smartphones and reliable internet connections.
- Creating a support network that includes follow-up training sessions and peer learning opportunities to ensure the long-term success of the program.

These workshops will enable women to independently access economic opportunities and improve their livelihoods.

D. Infrastructure Development

Strengthen digital infrastructure by installing community Wi-Fi hubs and providing devices in communal spaces. This will make digital access more consistent and reliable, particularly for women who lack personal access to devices at home.

E. Income-Generating Opportunities through Digital Platforms

Encourage and train women to use digital platforms for selling local products like handicrafts or agricultural goods. By teaching them to access wider markets, the village can foster greater economic independence for women.

F. Support and Collaboration with NGOs and Government Programs

Partner with NGOs and government agencies to ensure continued access to resources, funding, and additional skill-development programs. These partnerships could also support further infrastructure improvements and enhance training resources.

G. Encourage Community Engagement and Gender Inclusion

Promote a gender-inclusive environment where both men and women support and recognize the value of women's participation in digital and economic activities. This would involve community discussions and awareness programs to reduce gender biases and increase acceptance of women's digital empowerment.

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References

[1] S. Lythreatis, S. K. Singh, and A. N. El-Kassar, "The digital divide: A review and future research agenda," *Technol Forecast Soc Change*, vol. 175, 2022, doi: 10.1016/j.techfore.2021.121359.

[2] A. J. A. M. van Deursen and J. A. G. M. van Dijk, "The first-level digital divide shifts from inequalities in physical access to inequalities in material access," *New Media Soc*, vol. 21, no. 2, 2019, doi: 10.1177/1461444818797082.

[3] I. O. Adam and M. Dzang Alhassan, "Bridging the global digital divide through digital inclusion: the role of ICT access and ICT use," *Transforming Government: People, Process and Policy*, vol. 15, no. 4, pp. 580–596, 2020, doi: 10.1108/TG-06-2020-0114.

[4] L. Zhao, C. Cao, Y. Li, and Y. Li, "Determinants of the digital outcome divide in E-learning between rural and urban students: Empirical evidence from the COVID-19 pandemic based on capital theory," *Comput Human Behav*, vol. 130, May 2022, doi: 10.1016/j.chb.2021.107177.

[5] E. Yaacoub and M. S. Alouini, "A Key 6G Challenge and Opportunity-Connecting the Base of the Pyramid: A Survey on Rural Connectivity," *Proceedings of the IEEE*,

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vol. 108, no. 4, pp. 533–582, 2020, doi: 10.1109/JPROC.2020.2976703.

[6] "Ransai village." Accessed: Oct. 26, 2024. [Online]. Available: https://www.onefivenine.com/india/villages/Raigad/Khalapur/Ransai

[7] M. J. Bisheko and G. Rejikumar, "A study on farmers' perceptions about the scope of the Kisan Suvidha App in improving agricultural sustainability," in *2023 Conference on Information Communications Technology and Society, ICTAS 2023 - Proceedings*, 2023. doi: 10.1109/ICTAS56421.2023.10082741.

[8] "Ransai." Accessed: Oct. 26, 2024. [Online]. Available: https://mumbaimirror.indiatimes.com/ransai-an-environmental-delight/articleshow/15984471.cms

[9] A. Acilar and Ø. Sæbø, "Towards understanding the gender digital divide: a systematic literature review," 2023. doi: 10.1108/GKMC-09-2021-0147.

[10] GSMA, "The Mobile Gender Gap Report 2023," 2023.

[11] M. V. Ramesh, R. Mohan, and S. Menon, "Live-in-Labs: Rapid translational research and implementation-based program for rural development in India," in *GHTC 2016 - IEEE Global Humanitarian Technology Conference: Technology for the Benefit of Humanity, Conference Proceedings*, 2016. doi: 10.1109/GHTC.2016.7857275.

[12] R. Raman, P. Nedungadi, R. P. Amrita, S. Banerji, R. Mohan, and M. V. Ramesh, "Investigating the Factors Affecting the Adoption of Experiential Learning Programs: MBA Students Experience with Live-in-Labs," in *Proceedings of B-HTC 2020 - 1st IEEE Bangalore Humanitarian Technology Conference*, 2020. doi: 10.1109/B-HTC50970.2020.9297906.

[13] A. Prasad *et al.*, "Sustainable Technologies for Improved Income Generation: A Case Study of Women Villagers in Himachal Pradesh," in *Smart Innovation, Systems and Technologies*, 2022. doi: 10.1007/978-981-16-4177-0_65.

[14] O. Navya Teja, T. Srikar, G. Vinay Mahadev, D. Yaswanth, R. Mohan, and S. G. Divya Sharma, "Improving Income Generation Opportunities and Livelihood of Women in Pandori by Enhancing Their Skills," in *Lecture Notes in Electrical Engineering*, 2020. doi: 10.1007/978-981-15-1420-3_163.

[15] P. Sreeram *et al.*, *Technology Powered Resource Utilization for Income Generation Opportunities in the Rural Villages of Chhattisgarh*, vol. 601. 2020. doi: 10.1007/978-981-15-1420-3_176.

[16] M. V. Ramesh, A. Muir, K. Nandanan, R. R. Bhavani, and R. Mohan, "HCI Curricula for Sustainable Innovation the Humanitarian Focus at Amrita Vishwa Vidyapeetham," *Interactions*, vol. 29, no. 1, 2022, doi: 10.1145/3506564.

[17] K. G. V. N. Sai Kalyan *et al.*, "Women Empowerment through Productivity, Rural Development and Technology: Assessment of Cross Linkages within a Village of Maharashtra, India," in *IEEE Region 10 Humanitarian Technology Conference, R10-HTC*, 2020. doi: 10.1109/R10-HTC49770.2020.9356965.

[18] A. Neri *et al.*, "The role of digital technologies in supporting the implementation of circular economy practices by industrial small and medium enterprises," *Bus Strategy Environ*, vol. 32, no. 7, 2023, doi: 10.1002/bse.3388.

[19] US, "https://dhsprogram.com/publications/publication-FR375-DHS-Final-Reports.cfm," 2022.

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